An Ontology-Guided Approach for Enhanced Documentation of Cave Microbial Culture Collections

Christian Jewel M. Pascua and Mylah Rystie U. Anacleto

***Abstract*— Inconsistent documentation limits the scientific value of cave microbial data by making records harder to interpret and compare. Researchers using the 2023 CaveIS platform face this problem because it relies on free-text annotations that are prone to ambiguity and inconsistency. This study developed Caventology, a web application built on top of the existing cave culture collection information system, integrating ontology-based suggestions for selected annotation fields. It also provides an Ontology Lookup Service (OLS) to help users search for ontology terms, and an Ontology Explorer Service (OES) for examining their semantic relationships. Ontology-based suggestions and lookup functions were implemented using the BioPortal Search API, while RDF ontology files were parsed using N3.js and visualized with Cytoscape.js. Fifteen (15) survey participants composed of project stakeholders and researchers from science-related disciplines evaluated the system using the System Usability Scale, yielding a mean score of 79. Feedback and results show that Caventology offers a practical, ontology-guided approach to improving the consistency and semantic quality of cave microbial data.**

Index terms—ontology-guided documentation, cave microbiology, microbial culture collection, BioPortal, semantic web

I. Introduction

A. Background of the Study

Over the past decade, microbial research has gained renewed attention for its scientific and practical potential. From breaking down plastic [1] to supporting disease treatment [2], microorganisms are increasingly being studied to provide alternative solutions in emerging environmental and biomedical challenges. One such initiative, which this study is a part of, is the CAVE Project from the NICER CAVES Program, also known as the ”Phenomic, Genomic and Metagenomic Analysis of Bat Gut and Guano Microbiome from Caves in CALABARZON.” Led by the Microbial Culture Collection (MCC) research team from the University of the Philippines Los Banos Museum of Natural History (UPLB-MNH), its goal is to conduct a series of studies on bat gut and guano and to curate a collection of the datasets for all the sites studied.

Caves, which serve as the project’s primary focus, are geological formations appearing as voids that provide shelter to humans and animals [3]. Despite their seemingly inhospitable environment, they maintain a healthy ecosystem of their own, largely sustained by cave-dwelling bats. These bats are the main source of energy input from the outside world to the caves [4]. Their accumulated excrement, also called as guano, are responsible for nutrient cycling which help the cave support a diverse range of microbes that have piqued the interest of researchers [5].

The unique diet, habitat, and immune system of bats create the perfect conditions for specialized microorganisms to grow. Some of the microbes found in the gut and guano of bats show promising applications, such as lactic acid bacteria, that could be used as a probiotic [6]. However, the same conditions also allow harmful organisms such as the novel betacoronavirus to thrive [7]. Thus, there is a need for thorough documentation of cave microbes so the potential benefits and risks that they could bring are better understood. While the NICER CAVES Program addresses this by developing a cave culture collection information system, the systems being developed have yet to be refined to make them achieve the dataset quality and consistency of international microbial databases like BacDive and Omnicrobe.

This study addresses the limitations of CaveIS with an ontology-guided data entry workflow that promotes more consistent documentation of cave microbial data, and tools to improve users’ understanding of ontology terms and their relationships. **Ontologies**, which are a set of standardized terms following a defined structure [8], were used to provide suggestions in selected free-text annotations, reducing ambiguity during data entry. A graphical user interface (GUI) for an *Ontology Lookup Service (OLS)* and an *Ontology Explorer Service (OES)* were also provided to help users explore relevant ontology terms and better understand the relationships between the entities in each ontology.

B. Significance of the Study

The inception of CaveIS at the Microbial Culture Collection, Museum of Natural History, University of the Philippines Los Baños (UPLB) marks a major milestone in the institution’s efforts to document scientific information from cave sample collections. The building blocks for establishing a resource center for cave microorganism data have been laid by allowing researchers across the country to contribute and retrieve data collected from local caves. However, it must be noted that research transcends national boundaries; hence, the system’s microbial culture collection should ensure consistency of data to make them useful for broader scientific collaboration.

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This study enhances traditional cave microbial data documentation by leveraging ontology-based suggestions on selected free-text annotation fields–without drastically changing the data entry workflow established in CaveIS. Moreover, the system's *Ontology Lookup Service (OLS)* and *Ontology Explorer Service (OES)* allowed users to explore and better understand ontology terms so they could apply the system's suggestions more meaningfully. Together, these improvements bring the NICER CAVES Project closer to becoming a resource center for cave microorganism data, built for collaboration between local and international research institutions.

C. Objectives of the Study

This study sought to further develop CaveIS, which is a Culture Collection Information System for Cave Microorganisms developed in 2023.

To achieve this goal, the following objectives were accomplished:

1) to integrate ontology-based suggestions on free-text annotations to support more consistent data entry;
2) to develop a Graphical User Interface (GUI) for an *Ontology Lookup Service (OLS)* and an *Ontology Explorer Service (OES)* to help users explore relevant ontology classes and their relationships; and
3) to evaluate the usability of the system to be developed using the System Usability Scale (SUS).

D. Scope and Limitations

Since the cave culture collection information system was conceived to cater to researchers, its users are limited to individuals who have the capacity to understand technical information in Biology. Moreover, accessing the information system is limited to web browsers only and it has no support for offline services. The functionality for providing ontology-based suggestions is limited to selected free-text annotation fields when adding isolate entries. However, users need not to select from these ontology-based suggestions, as they are only provided as guides and not as required input options.

II. Related Literature

The success of international biodiversity databases has inspired the NICER CAVES Program to envision a local implementation of a public online database for cave culture research. A number of projects have been developed to fulfill this goal, however, not all of them are robust enough to cater to the precise demands of scientific research. In this review of related literature, the theoretical and technical foundations behind the integration of ontology in microbial data documentation will be explored. This section will discuss how the semantic web spurred further incorporation of ontologies in public online databases. In addition, the theoretical groundwork for representing ontologies in web-based systems will be assessed for understanding the potential of advanced information systems in cave microbial research.

A. Ontology and its Applications in Life Sciences

Ontology is a concept originating from the philosophical study of being and existence [9]. In Edmund Husserl's "Logical Investigations," he distinguishes formal ontology as the study of describing objects and the relationships between them. Its ultimate goal is to craft a universally-applicable language for defining and categorizing knowledge. Ontologies have existed in the biomedical domain for as early as the 1700s. The basis of our modern taxonomy and understanding of evolution can be traced back to the work of Charles Darwin on the Linnaean taxonomy which describes relationships between species [9]. Nowadays, ontologies have found its uses beyond taxonomic classifications, becoming an essential part of modern biomedical sciences. One of the oldest and most widely adopted is the Gene Ontology (GO) which is used in describing molecular functions of gene products from organisms using standardized language [10].

Since the success of GO, several ontologies have been developed to suit the needs of the different domains under the life sciences [11]. The National Center for Biotechnology Information (NCBI) Organismal Classification (NCBITAXON) is now the de facto standard for organism names and taxonomic lineages [12]. For describing phenotypic qualities, there is the Phenotype And Trait Ontology (PATO) [13]. Some ontologies combine multiple ontologies that are useful to their domain. This applies to the Ontology for Biomedical Investigations (OBI) which imports terms from GO and PATO that are relevant for life-science and clinical investigations [14]. On the other hand, there are sets of ontologies that describe mostly the same entities but have diverging approaches on how to describe and categorize them. Such is the case with Environment Ontology (ENVO) and Microbial Isolation Source Ontology (MISO) which target environments. The difference lies in ENVO focusing on the high-level categorization of the environment [15] while MISO focuses more on the context of the environment from where the sample was taken [16].

B. Parsing and Visualizing Ontologies in Web-Based Systems

In the context of bioinformatics, **ontologies** are defined as "relationship-based hierarchical descriptions of concepts within a particular domain, such as biomedicine." [17] They consist of entities in a tree or acyclic graph structure connected via "is-a", "part-of" and other semantic relationships [18]. Generally, they are distributed in machine-readable formats such as OWL (Web Ontology Language) and RDF (Resource Description Framework). Since these file formats are relatively obscure, applications that wish to utilize them require specific methods for parsing and visualization.

In web-based applications, there are Javascript libraries that can process ontology file formats. One of these is the *N3.js* library which provides support for parsing and writing RDF data [19]. It transforms the parsed data into triples, which define a subject, predicate, and object; or into quads, which extend triples by adding a metadata component. These structures can then be further processed for search, analysis, or visualization. For representing the parsed data interactively in the browser, the *Cytoscape.js* library is particularly useful since it can visualize semantic relationships between ontology classes into node-edge visualizations. [20]

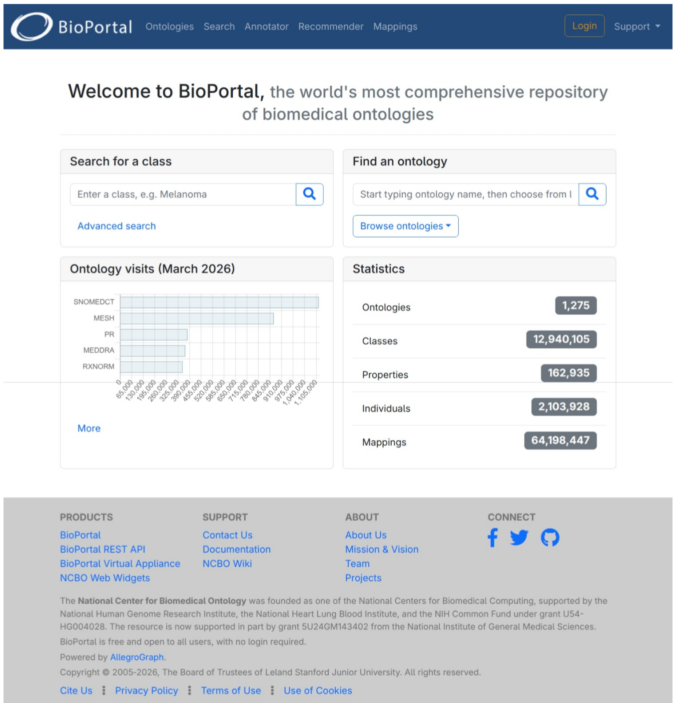

Fig. 1. The National Center for Biomedical Ontology (NCBO) BioPortal Homepage

Several studies foresee ontologies being integrated into more and more biodiversity databases in the future [21] [22], making access to ontology content become increasingly essential. In line with this, the National Center for Biomedical Ontology (NCBO) developed BioPortal, which provides web services that allow software applications to access ontology metadata, classes, and search functionality [23]. It has API endpoints for ontology term search, annotator, and recommender, making it useful for systems that need ontology lookup and suggestion features during data entry.

C. The Semantic Web as a Catalyst for Public Online Databases

There may be a multitude of reasons why ontologies have been created or forked from existing ones, but all of them were conceived with the idea of making vast amounts of data standardized and machine-readable, aligning with the primary goals of the Semantic Web. The Semantic Web provides tools to structure and organize heterogeneous information through sets of standards and technologies [24]. It was first coined by Tim Berners-Lee in his paper, ”Publishing on the Semantic Web” [25] after foreseeing an explosion of information with the emergence of the Internet. Due to the fragmented, multi-dimensional nature of properties and attributes characterizing biomedical phenomena, many life-sciences researchers have been early adopters of the movement [24]. The strong cooperation of the biomedical field with the Semantic Web has birthed the **W3C Web Ontology Language (OWL)** which further catalyzed the creation of ontologies in the biomedical domain.

Consequently, efforts to consolidate and standardize microbial metadata have facilitated the development of biomedical information systems which are public online databases that have become the premier knowledge repositories of researchers in the field today. For instance, the Bacterial Diversity Metadatabase of Leibniz Institute DSMZ also known as BacDive, is now the flagship database for standardized prokaryotic information on strain level with ”82 892 bacterial and archaeal strains covering taxonomy, morphology, cultivation, metabolism, origin, and sequence information within 1 048 data fields” as of July 2021 [16]. The BacDive web service also offers RESTful APIS for large-scale data retrieval and analyses through the URL: `https://api.bacdive.dsmz.de/`. On the flip side, the Genomes OnLine Database (GOLD) from the Department of Energy Joint Genome Institute (DOE-JGI) is one of the leading genomic metadata repositories in the world with over 54 052 studies and 468 058 GOLD organisms in its database [26]. The resounding success of these public online databases is what the NICER CAVES Program aims to replicate, albeit on a smaller scale, as it seeks to preserve the vast amounts of cave microorganism data produced by research institutions all over the country and make it accessible not only at the national but global level.

III. Methodology

A. Development Tools

The **Caventology** cave culture collection information system was developed and tested on a laptop with the following specifications:

- Operating System: Ubuntu 25.10 (Questing Quokka)
- Processor: AMD Ryzen 5 4500 U CPU @ 2.38 GHz, 6 Core(s), 12 Logical Processor(s)
- Memory: 16 GB DDR4 @ 3200 MHz

Furthermore, the development of the system used a suite of tools and technologies which include but are not limited to:

- **Visual Studio Code** - a source-code editor made by Microsoft which served as the main Integrated Development Environment (IDE) for the project.
- **MySQL** - a free and open-source database management system by Oracle which was used to implement the database.
- **Django** - a Python web application framework that was used for the system's backend.
- **React** - a free and open-source JavaScript library used to build the system's user interface.
- **Material UI** - an open-source library of Material design implementations of React components, utilized for the system's user interface.
- **N3.js** - a JavaScript library which was used to parse RDF (Resource Description Framework) data inside the browser environment.

- **Cytoscape.js** - an open-source graph-theory JavaScript library which was used to visualize node-edge relationships among ontology entities.

B. User Levels and Privileges

The system provides three user types with varying levels of privileges: Guest, Researcher, and Administrator. The Guest user type can use the OLS and OES tools and view isolate records with public access level while the Researcher user type can also manage source and isolate records in the system's database. Meanwhile, the Administrator has extended capabilities to manage the system's user accounts. A comprehensive list of the privileges for each user type are enumerated on the next page:

1) Guest
   - Guests can view isolate records with *public* access level.
   - Guests can explore ontology entities in the Ontology Explorer Service (OES).
   - Guests can look up ontology terms in the Ontology Lookup Service (OLS).
2) Researcher
   - Researchers can view isolate records with *public* and *private* access levels.
   - Researchers can bulk import isolate records.
   - Researchers can create and delete isolate records.
   - Researchers can set the access levels of each isolate record.
   - Researchers can explore ontology entities in the Ontology Explorer Service (OES).
   - Researchers can look up ontology terms in the Ontology Lookup Service (OLS).
3) Administrator
   - Administrators can view isolate records with *public* and *private* access levels.
   - Administrators can bulk import isolate records.
   - Administrators can create and delete isolate records.
   - Administrators can set the access levels of each isolate record.
   - Administrators can explore ontology entities in the Ontology Explorer Service (OES).
   - Administrators can look up ontology terms in the Ontology Lookup Service (OLS).
   - Administrators can create and delete Guest, Researcher, and Administrator user types.
   - Administrators can view the activity logs created by all users such as creating and deleting isolate records and user account details.

C. Application Features

The **Caventology** web application can be accessed through a web browser with access to the Internet. It has features for User Account Management, Isolate Records Management, an Ontology Lookup Service (OLS) and an Ontology Explorer Service (OES). A more comprehensive breakdown of the features are listed afterwards.

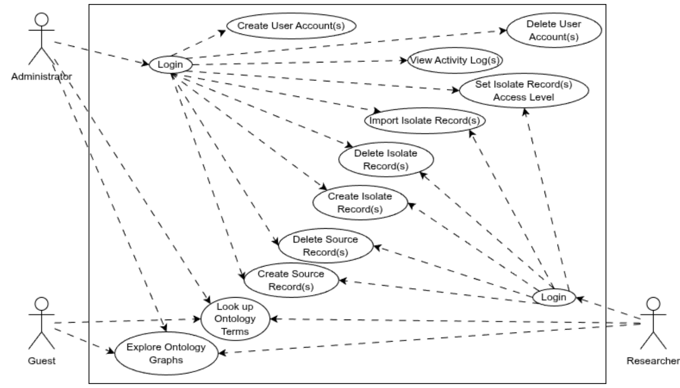

Fig. 2. Use Case Diagram for the Caventology Web App

- User Account Management
  1) Create User Account
  2) Delete User Account
  3) Login
  4) Logout
  5) View Database Activity Log(s)
- Source Records Management
  1) Create Source Record
  2) Delete Source Record
  3) Create Source Record Metadata
  4) Delete Source Record Metadata
  5) View Source Record(s)
  6) Filter Source Record(s) by ID
  7) Filter Source Record(s) by Host Species
- Isolate Records Management
  1) Create Isolate Record
  2) Delete Isolate Record
  3) Create Isolate Record Metadata
  4) Delete Isolate Record Metadata
  5) Import Isolate Record(s)
  6) View Isolate Record(s)
  7) Filter Isolate Record(s) by ID
  8) Filter Isolate Record(s) by Accession Number
- Ontology Lookup Service (OLS)
  1) Search Ontology Term(s)
  2) Filter Ontology Term Search result(s) by Specific Ontology
  3) View Ontology Term Metadata
  4) View Ontology Term Metadata on BioPortal
- Ontology Explorer Service (OES)
  1) Import RDF File
  2) Visualize Ontology Term(s) using Hierarchical Layout
  3) Visualize Ontology Term(s) using Circular Layout
  4) Visualize Ontology Term(s) using Grid Layout
  5) Visualize Ontology Term(s) using Force-directed (CoSE) Layout
  6) Visualize Ontology Term(s) using Breadth-First Layout
  7) View Ontology Term Metadata

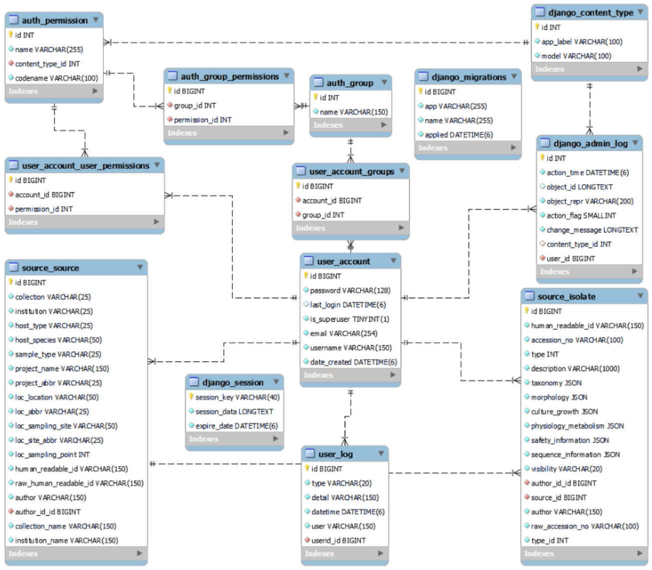

Fig. 3. Database Enhanced Entity Relationship Diagram (EERD) for the Metadata Standards Established by Other CAVE Program Researchers [27]

D. Database Design

The database design for the Caventology web application uses three tables: User Table, Source Table, and Isolates Table. The tables for User, Source, and Isolates follow the metadata standards established by other CAVE project researchers. The Enhanced Entity Relationship Diagram (EERD) for the three tables is shown in Figure 3.

IV. Results and Discussion

The Caventology web application was primarily built using *React* for the client side, *Django* for the server side, and *MySQL* for the database layer. Its user interface was developed with *Material UI*, which provide an extensive collection of Material Design implementations of various web user interface elements. The library's pre-built components and ability to define custom components helped expedite the development stage while still ensuring the web app is user-friendly and web standards-compliant.

Since the web application builds upon the existing CaveIS architecture, the server side of the app uses the API endpoints and database system used in CaveIS. However, the ontology based-suggestions and the *Ontology Lookup Service (OLS)* of the app fetches data hosted in BioPortal through its Search API. BioPortal provides a free API key enabling access to over 12 million classes from several ontologies hosted in their website. On the other hand, the *Ontology Explorer Service (OES)* processes RDF files through the use of *N3.js parser*. It then stores the parsed data as quads in the memory which are further processed into metadata, nodes, and edges arrays. The arrays are used by *Cytoscape.js* to visualize them into an interactive graph, with several layout options to choose from.

A. Web Application

1) **Login Page**
The login page provides a portal for the users with existing accounts to log in to the system with their email and password credentials. Unauthenticated users are redirected to this page whenever they try to access services that are not within their user privileges.

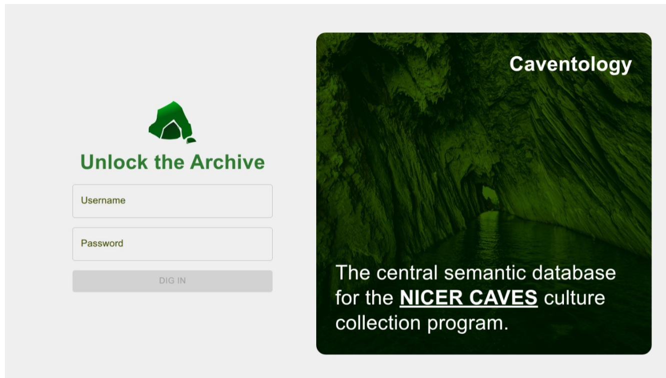

Fig. 4. Caventology Login Page

2) **Home Page**
On the home page, users can view the total number of source and isolate records in the overall database. There is also a short explainer about ontologies and its use case for cave microbial research. The two main features for learning more about ontologies are highlighted on the home page with quick links to access them.

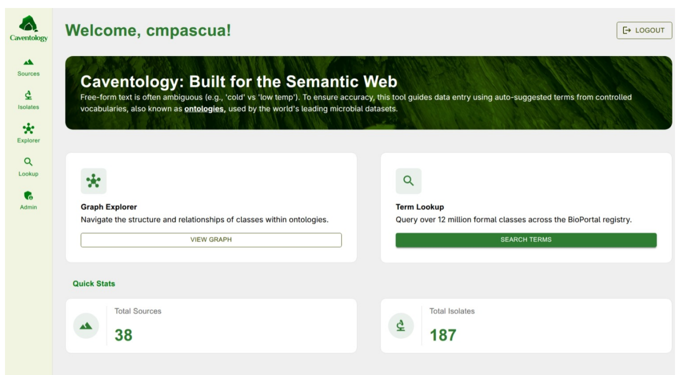

Fig. 5. Caventology Home Page

3) **Sources Page**
The sources page lists down all the source records visible to the current user. It also provides the functionality to filter source records by ID and/or host species. Users with sufficient privileges can also delete a source record by clicking the "Delete" button on a selected row.

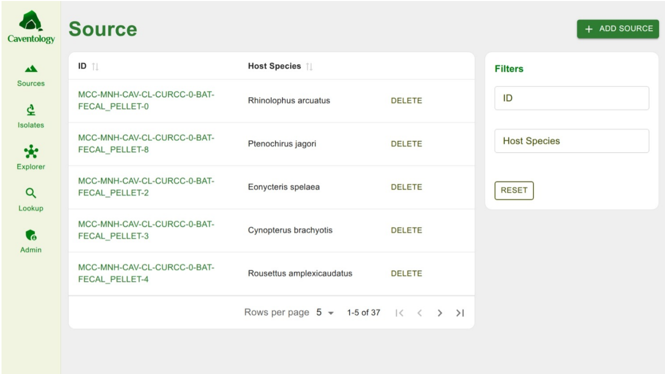

Fig. 6. Caventology Sources Page

Clicking the ID hyperlink on a source record brings the user to the source metadata page, which lists down all available metadata of the source record. The isolate records associated with the selected source record are also listed down on the right side of the page.

Fig. 7. Caventology Source Details Page

Clicking the "Add Source" button enables the user to add a new source record with the following metadata options:

- Source Information
  - Collection Name
  - Institution Name
  - Project Name
- Host Information
  - Host Type
  - Host Species
  - Sample Type
- Sampling Information
  - Location
  - Sampling Site
  - Sampling Point

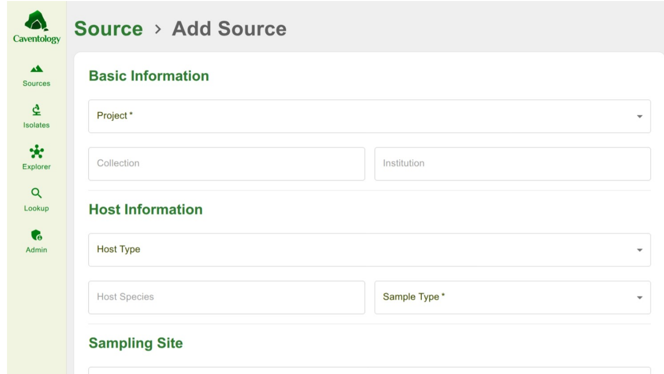

Fig. 8. Caventology Add Source Page

4) **Isolates Page**
The isolates page lists down all the isolate records visible to the current user. It also provides the functionality to filter isolate records by ID and/or accession number. Users with sufficient privileges can also delete an isolate record by clicking the "Delete" button on a selected row.

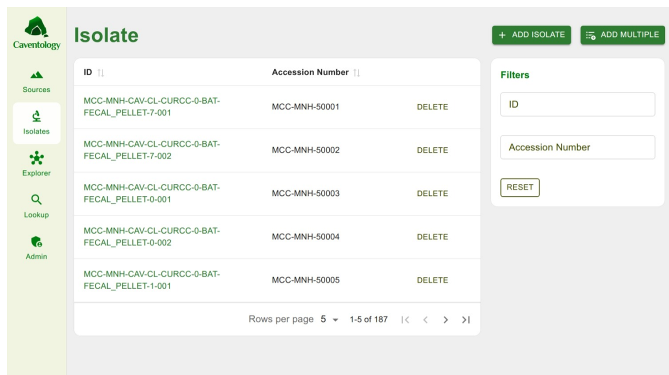

Fig. 9. Caventology Isolates Page

Clicking the ID hyperlink on an isolate record brings the user to the isolate metadata page, which lists down all available metadata of the isolate record. The source record associated with the selected isolate record is also listed on the right side of the page. To import multiple isolate records, users can click on the "Add Multiple" button which lets them select a CSV file to bulk import records.

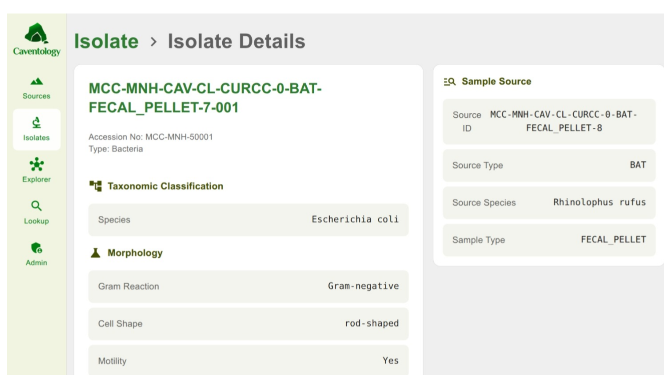

Fig. 10. Caventology Isolate Details Page

On the other hand, clicking the "Add Isolate" button enables the user to add a new isolate record with the following metadata options, providing ontology-based suggestions on selected fields with an asterisk:

- Source Information
  - Source ID
  - Isolate Type
- Taxonomic Classification (optional)
  - Domain*
  - Phylum*
  - Class*
  - Order*
  - Family*
  - Genus*
  - Species*
- Morphology (optional)
  - Gram Reaction
  - Cell Shape
  - Motility
- Culture and Growth Conditions (optional)
  - Culture Medium*
  - Growth
  - Culture Medium Composition
  - Growth Temperature
  - Temperature Range
- Physiology and Metabolism (optional)
  - Oxygen Requirement*
  - Presence of Cytochrome C Oxidase
  - Endospore Forming Capability
  - Antibiotic Resistance Profile*
- Safety Information (optional)
  - Pathogenicity (Human)
  - Pathogenicity (Animals)
  - Biosafety Level
- Visibility

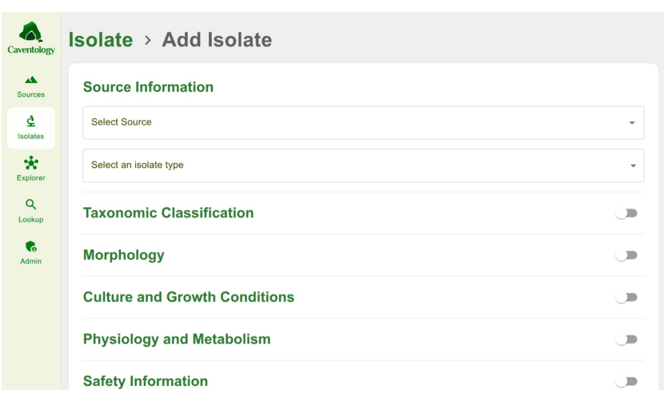

Fig. 11. Caventology Add Isolate Page

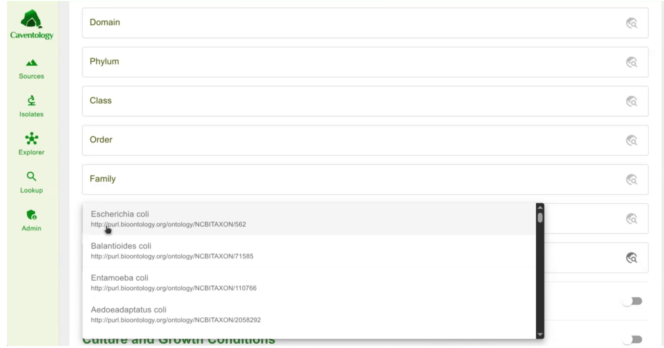

Fig. 12. Caventology Add Isolate Page - Ontology Suggestions

5) **OLS Page**
On the OLS page, users can access over 12 million ontology classes hosted in BioPortal with the lookup feature. Users can look up relevant ontology terms by typing in the search box. They can also look for ontology terms in specific ontologies through the filter options.

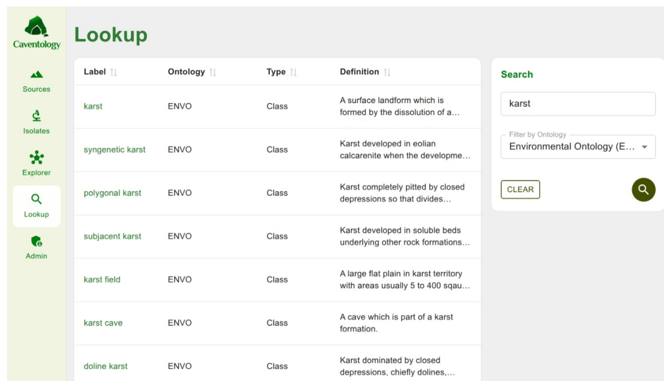

Fig. 13. Caventology OLS Page

Clicking the label hyperlink for each search result takes the user to the Ontology Details page which provides metadata for definition(s), identifier(s), and related ontology terms (if there are any). Users can click the "View in BioPortal" button to view more information about the selected ontology class in BioPortal.

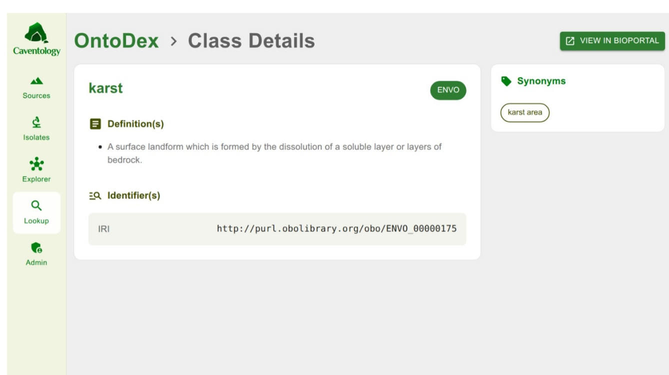

Fig. 14. Caventology OLS Details Page

6) **OES Page**
The OES page brings out an interactive visualization of ontologies. Users can click the ”Import File” button to import RDF data. The system parses the file with *N3.js* parser and then it uses the help of *Cytoscape.js* to visualize ontology classes into node and edge relationships. The metadata of the ontology is located on the right panel of the page. It displays the total number of nodes and edges in the ontology and the ontology description. Clicking a node on the page displays the label, type, and the uniform resource identifier (URI) of the selected ontology class.

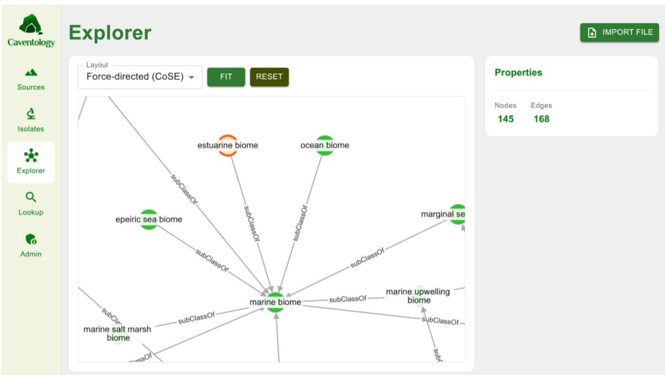

Fig. 15. Caventology OES Page

7) **Admin Page**
The Admin page provides views for the activity log and all existing user accounts of the system for the Administrator user type. The activity log lists down the action, the date and time of the action, and the user that executed the action on the system. The user accounts view lists down the username and email of each user with an option to delete them through the ”Delete” button. Should the administrator wish to add a user, they can click on the ”Add Account” button which pops up a modal page with username, email address, and password fields to add a new user account.

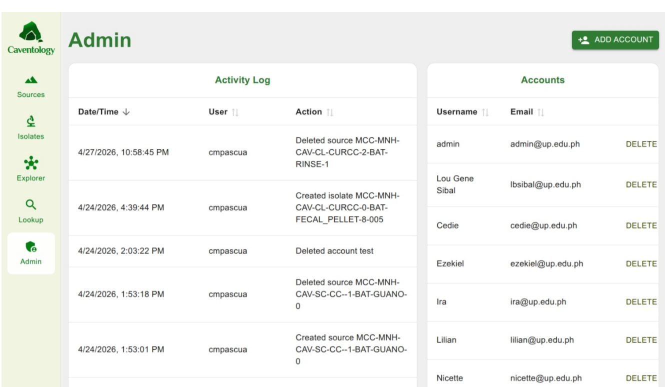

Fig. 16. Caventology Admin Page

B. Evaluation

The Caventology web application’s usability and effectiveness were evaluated using the System Usability Scale (SUS). The respondents consisted of a diverse group of 15 researchers and faculty under the NICER Program: Center for Assessment of Cave Natural Resources (CAVE) in CALABARZON, affiliated with the Microbial Culture Collection, Museum of Natural History, UPLB; and students under the BS Biology, Agricultural Biotechnology (ABT), and Math and Science Teaching (MST) degree programs.

Each participant were given a briefer for the use of ontologies in a database management system and were given detailed instructions on how to navigate the web application. The participants underwent scenarios which made them browse through all the features of the app, evaluating its ease of use and user interface.

The participants were then tasked to answer an SUS questionnaire on paper after using the web application. SUS is a 10-question framework using the Likert scale which consists of the following questions:

1) I think that I would like to use this system frequently.
2) I found the system unnecessarily complex.
3) I thought the system was easy to use.
4) I think that I would need the support of a technical person to be able to use this system.
5) I found the various functions in this system were well integrated.
6) I thought there was too much inconsistency in this system.
7) I would imagine that most people would learn to use this system very quickly.
8) I found the system very cumbersome to use.
9) I felt very confident using the system.
10) I needed to learn a lot of things before I could get going with this system.

TABLE I
SYSTEM USABILITY SCALE (SUS) RESULTS

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Score
R1	5	1	5	1	5	1	3	1	5	1	95.0
R2	5	1	5	1	5	1	5	1	5	1	100.0
R3	5	1	5	5	5	1	5	1	5	5	80.0
R4	5	5	5	5	5	5	5	1	5	4	62.5
R5	4	3	3	1	4	2	5	2	4	3	72.5
R6	5	2	4	2	5	1	4	2	4	3	80.0
R7	4	1	3	2	3	2	3	2	3	1	70.0
R8	3	2	4	2	3	2	3	2	3	2	65.0
R9	5	1	5	3	5	1	5	1	5	1	95.0
R10	4	2	4	2	4	1	4	2	3	2	75.0
R11	5	1	5	2	4	1	4	1	5	2	90.0
R12	4	2	4	2	3	3	5	1	4	2	75.0
R13	4	2	4	2	4	2	4	2	4	2	75.0
R14	5	1	4	2	5	2	4	1	4	2	85.0
R15	4	3	3	3	4	1	5	2	3	4	65.0
Mean											**79.0**

The threshold for above-average system usability is 68. Based on the participant ratings shown in the figure, Caventology obtained a System Usability Scale score of **79**, corresponding to a ”Good” adjective rating. While conducting the survey, the respondents also provided comments beyond the scope of the questionnaire. Most respondents remarked that Caventology was visually appealing and easy to navigate. The BS Biology students, in particular, appreciated the tight integration of the application's features. They mentioned that research in biology is often citation-heavy, requiring them to manage multiple browser tabs during the research process. Having an app that consolidates all the features they need reduces their dependence on switching between a lot of browser tabs and windows.

On the flip side, there are specific areas where improvements could be made to enhance the app's usability. One respondent mentioned the lack of confirmation dialogs on critical system operations such as deleting user accounts and deleting source and isolate records. Another respondent commented that it would be more helpful if the view for the *OES* would only include a snapshot of the ontology if it is too large to avoid overwhelming the user. Most ontologies contain hundreds or thousands of ontologies and visualizing them all at once (like the current system does) would make them very small to see and hard to navigate. An approach that would only load sibling and parent classes when clicking would be useful to improve the navigability of the *OES*.

V. Conclusion

This study developed **Caventology** with an ontology-guided approach for improving the documentation of cave microbial culture collections. The integration of ontology-based suggestions on free-text annotation fields helped users achieve more consistency within microbial records. The *OLS* provided user access to over 12 million ontology terms from the BioPortal repository, while the *OES* allowed users to explore relationships among ontology classes. These features supported a more systematic approach to data entry and gave users resources for understanding ontology terms.

Furthermore, the System Usability Survey showed that Caventology has above average usability for its use-case. Garnering an SUS score of **79** and mostly positive feedback from survey participants, the system demonstrated its effectiveness as a tool for ontology-guided documentation. Given these, the study has achieved its objectives and demonstrated the value of ontology integration in improving consistency, semantic quality, and research value of cave microbial records.

VI. Recommendation

There lies a lot of opportunities for future researchers to improve the web application. First, stronger interoperability among its features could be developed. In its current version, the *Source* and *Isolate Records Management*, *OLS*, and *OES* features are functionally and architecturally separate. For instance, viewing the definition of a suggested term while adding an isolate record requires the user to leave the *Add Isolate Record* page and open the *OLS* page, disrupting the workflow. In a similar vein, viewing additional information about specific nodes in the *OES* page is only possible through the *OLS* page. Future versions of the system could therefore integrate these features more closely so users can access relevant ontology information without interrupting their current task.

On the other hand, the ranking and relevance of ontology-based suggestions could also be improved. Instead of fetching data directly from the BioPortal Search API, the system could further process the retrieved terms to generate improved suggestions. Following that, it would also be useful to extend ontology-based suggestions to more metadata fields, if not all fields, in the system. However, a more ambitious direction would be the development of a cave microbe ontology tailored specifically to the needs of cave microbial research. Such an ontology would reduce the system's dependence on the BioPortal API, support faster feature development, and further improve consistency in data entry. If achieved, this contribution could provide a valuable resource for the cave microbe research community and mark an important milestone for the NICER CAVES project.

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References

[1] S. K. S. Patel and J.-K. Lee, "Plastic Eating Enzymes: A Step Towards Sustainability," Indian Journal of Microbiology, vol. 62, no. 4, pp. 658–661, Sep. 2022, doi: https://doi.org/10.1007/s12088-022-01041-w.

[2] Z. Winn, "Turning microbiome research into a force for health," MIT News — Massachusetts Institute of Technology, Jan. 05, 2021. https://news.mit.edu/2021/microbiome-research-health-0105.

[3] S. Alcazar et al., "Cave bats and their habitats in Nug-as and Mt. Lantoy key biodiversity area (KBA), Cebu, Philippines," vol. 25, pp. 621–637, Jan. 2016.

[4] IUCN SSC, Ed., "IUCN SSC Guidelines for Minimizing the Negative Impact to Bats and Other Cave Organisms from Guano Harvesting," Mar. 2014. Available: https://portals.iucn.org/library/sites/library/files/documents/Rep-2014-002.pdf.

[5] M. P. De Leon, A-young. Park, A. D. Montecillo, M. A. T. Siringan, A. R. R. Rosana, and S.-G. Kim, "Near-Complete Genome Sequences of Streptomyces sp. Strains AC1-42T and AC1-42W, Isolated from Bat Guano from Cabalyorisa Cave, Mabini, Pangasinan, Philippines," Microbiology Resource Announcements, vol. 7, no. 7, Aug. 2018, doi: https://doi.org/10.1128/mra.00904-18.

[6] Souraya Sakoui et al., "The first study of probiotic properties and biological activities of lactic acid bacteria isolated from Bat guano from Er-rachidia, Morocco," Lebensmittel-Wissenschaft + Technologie/Food science & technology, vol. 159, pp. 113224–113224, Apr. 2022, doi: https://doi.org/10.1016/j.lwt.2022.113224.

[7] Pawel Fedurek et al., "Selective deforestation and exposure of African wildlife to bat-borne viruses," Communications biology, vol. 7, no. 1, Apr. 2024, doi: https://doi.org/10.1038/s42003-024-06139-z.

[8] K. Dumschott et al., "Ontologies for increasing the FAIRness of plant research data," Front. Plant Sci., vol. 14, Nov. 2023, doi: 10.3389/fpls.2023.1279694.

[9] L. J. Jensen and P. Bork, "Ontologies in Quantitative Biology: A Basis for Comparison, Integration, and Discovery," PLoS Biology, vol. 8, no. 5, p. e1000374, May 2010, doi: https://doi.org/10.1371/journal.pbio.1000374.

[10] M. Ashburner et al., "Gene Ontology: tool for the unification of biology," Nature Genetics, vol. 25, no. 1, pp. 25–29, May 2000, doi: https://doi.org/10.1038/75556.

[11] D. L. Rubin, N. H. Shah, and N. F. Noy, "Biomedical ontologies: a functional perspective," Briefings in Bioinformatics, vol. 9, no. 1, pp. 75–90, Oct. 2007, doi: https://doi.org/10.1093/bib/bbm059.

[12] S. Federhen, "The NCBI Taxonomy database," Nucleic Acids Research, vol. 40, no. D1, pp. D136–D143, Dec. 2011, doi: https://doi.org/10.1093/nar/gkr1178.

[13] G. V. Gkoutos, P. N. Schofield, and R. Hoehndorf, "The anatomy of phenotype ontologies: principles, properties and applications," Briefings in Bioinformatics, vol. 19, no. 5, pp. 1008–1021, Apr. 2017, doi: https://doi.org/10.1093/bib/bbx035.

[14] A. Bandrowski et al., "The Ontology for Biomedical Investigations," PLOS ONE, vol. 11, no. 4, p. e0154556, Apr. 2016, doi: https://doi.org/10.1371/journal.pone.0154556.

[15] Pier Luigi Buttigieg, Evangelos Pafilis, S. E. Lewis, M. Schildhauer, R. Walls, and C. J. Mungall, "The environment ontology in 2016: bridging domains with increased scope, semantic density, and interoperation," Journal of Biomedical Semantics, vol. 7, no. 1, Sep. 2016, doi: https://doi.org/10.1186/s13326-016-0097-6.

[16] L. C. Reimer, J. Sardà Carbasse, J. Koblitz, C. Ebeling, A. Podstawka, and J. Overmann, "BacDive in 2022: the knowledge base for standardized bacterial and archaeal data," Nucleic Acids Research, vol. 50, no. D1, pp. D741–D746, Oct. 2021, doi: https://doi.org/10.1093/nar/gkab961.

[17] T. Nix, "Research Guides: Biomedical Ontologies and Controlled Vocabularies: Overview." Accessed: Nov. 22, 2024. [Online]. Available: https://guides.lib.umich.edu/ontology/overview

[18] A. Groß, C. Pruski, and E. Rahm, "Evolution of biomedical ontologies and mappings: Overview of recent approaches," Computational and Structural Biotechnology Journal, vol. 14, pp. 333–340, Jan. 2016, doi: 10.1016/j.csbj.2016.08.002.

[19] R. Verborgh et al., "rdfjs/n3.js: v2.0.3," Zenodo, Mar. 2024, doi: 10.5281/zenodo.10866356.

[20] M. Franz, C. T. Lopes, G. Huck, Y. Dong, O. Sumer, and G. D. Bader, "Cytoscape.js: a graph theory library for visualisation and analysis," Bioinformatics, vol. 32, no. 2, pp. 309–311, Sep. 2015, doi: 10.1093/bioinformatics/btv557.

[21] S. Trißl and U. Leser, "Querying Ontologies in Relational Database Systems," Lecture Notes in Computer Science, pp. 63–79, 2005, doi: https://doi.org/10.1007/11530084_7.

[22] N. Laurenne, J. Tuominen, Hannu Saarenmaa, and Eero Hyvönen, "Making species checklists understandable to machines – a shift from relational databases to ontologies," Journal of Biomedical Semantics, vol. 5, no. 1, pp. 40–40, Jan. 2014, doi: https://doi.org/10.1186/2041-1480-5-40.

[23] P. L. Whetzel et al., "BioPortal: enhanced functionality via new Web services from the National Center for Biomedical Ontology to access and use ontologies in software applications," Nucleic Acids Research, vol. 39, no. suppl, pp. W541–W545, Jun. 2011, doi: 10.1093/nar/gkr469.

[24] A. Splendiani, A. Burger, A. Paschke, P. Romano, and M. S. Marshall, "Semantic Web, Ontology, and Linked Data," https://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/978-1-5225-1741-2.ch009. Accessed: Nov. 22, 2024. [Online]. Available: https://www.igi-global.com/gateway/chapter/www.igi-global.com/gateway/chapter/174331.

[25] T. Berners-Lee and J. Hendler, "Publishing on the semantic web," Nature, vol. 410, no. 6832, pp. 1023–1024, Apr. 2001, doi: https://doi.org/10.1038/35074206.

[26] S. Mukherjee et al., "Twenty-five years of Genomes OnLine Database (GOLD): data updates and new features in v.9," Nucleic Acids Research, vol. 51, no. D1, pp. D957–D963, Jan. 2023, doi: 10.1093/nar/gkac974.

[27] N. H. Garcia, A. Jacildo, and M. R. Anacleto, "Cave Microbiome Database: A Database System for Cave Microorganism Data in the Philippines".

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